

# **AI-POWERED DRONE SYSTEM FOR COCONUT DISEASE DETECTION, REAL-TIME MONITORING AND INTERVENTIONS**

Project ID: 25-26J-384

Project Proposal Report

B.Sc(Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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# Declaration

## DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

| Name               | Student ID | Signature                                                                          |
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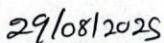
The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Name of supervisor: Ms.Uthpala Samarakoon

  
.....

Signature of the supervisor:

(Ms.Uthpala Samarakoon)

  
.....

Date

## Abstract

Coconut is a vital plantation crop in Sri Lanka, contributing significantly to the national economy through local consumption and export-based industries. However, accurately forecasting coconut yield and determining the maturity stages of coconuts classified as early, middle-aged, and brown remains a manual, time-consuming, and often inaccurate task. In traditional farming practices, decisions regarding harvest timing and yield estimation are largely based on human observation, which lacks consistency and scalability. These limitations reduce efficiency in crop planning, resource allocation, and disease intervention strategies.

This research proposes the development of an intelligent drone-assisted system powered by deep learning techniques to automate yield prediction and coconut maturity classification. Drone imagery will be captured from coconut plantations and analyzed using computer vision techniques. Convolutional Neural Networks (CNNs) will be applied to detect and classify coconuts by maturity stage based on image features such as color, texture, and size. For yield prediction, statistical regression models and deep learning algorithms will be employed using variables such as crown size, nut count, and vegetation indices derived from drone images.

The proposed system is designed to empower coconut farmers and agricultural officers with real-time, data-driven insights, enabling better harvesting decisions and improved yield estimation. This automation aims to reduce manual effort, enhance accuracy, and provide early planning capabilities for resource management. Integrating this component into the broader AI-powered drone platform will further support disease detection, monitoring, and intervention in coconut farming.

**Keywords:** Coconut Yield Prediction, Maturity Classification, Drone Imagery, Convolutional Neural Networks (CNNs), Deep Learning, Computer Vision, Smart Agriculture

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# 1. Introduction

Coconut cultivation is a cornerstone of Sri Lanka's agricultural economy, contributing to both domestic consumption and international exports. As one of the country's most widely grown plantation crops, the coconut palm supports the livelihoods of thousands of rural families. However, maximizing coconut productivity is still a challenge due to factors such as irregular harvesting, disease spread, climate variation, and inefficient monitoring techniques. Traditional methods of estimating yield and assessing the maturity of coconuts rely heavily on manual observation, which can be time-consuming, inconsistent, and prone to error.

Recent advances in artificial intelligence (AI), drone imaging, and computer vision technologies offer promising solutions to modernize coconut farming practices. Drone-based image collection enables scalable, non-invasive, and real-time monitoring of coconut plantations. AI-powered image processing can be used to automatically identify and classify coconuts based on maturity stages (early, middle-aged, brown) using visual cues such as color and size. Furthermore, AI models can be trained to predict the yield per palm or per plot based on image features such as crown size, nut count, and vegetative health.

The classification of coconuts based on their maturity stage plays a critical role in determining the optimal harvest time. Early harvesting may reduce oil content, while late harvesting risks over-ripeness and disease exposure. Hence, accurate classification helps optimize harvest scheduling, storage, and processing. Similarly, yield prediction allows farmers and cooperatives to forecast expected production levels, allocate resources effectively, and plan for market distribution in advance.

This research component aims to design and implement an intelligent yield prediction and coconut maturity classification system using drone-collected imagery and deep learning algorithms. By integrating this system into the broader AI-powered drone framework, farmers will gain access to automated insights that will enhance planning, reduce labor dependency, and improve overall efficiency in coconut farming. This effort aligns with the growing need for digital transformation in agriculture and supports the vision of sustainable, data-driven plantation management in Sri Lanka.



Figure 1.1: Immature (Green) Coconuts on Tree



Figure 1.2: Coconut Trees with Mixed Maturity Fruits

## 1.1 Background and Literature Survey

Coconut cultivation is a cornerstone of Sri Lanka's agriculture industry, significantly contributing to the national economy and rural livelihoods. With over 394,000 hectares of coconut plantations across the country, coconuts are not only a primary export crop but also a staple in local consumption. The coconut palm provides various by-products such as oil, fiber, husk, and coir, making it a highly valuable resource. However, the yield and quality of coconuts are heavily influenced by factors such as pests, diseases, nutrient levels, climate variability, and harvesting practices.

Traditionally, farmers estimate coconut yield and classify coconuts by maturity stage manually. This process is labor-intensive, time-consuming, and highly subjective, which may lead to inconsistent yield predictions and improper categorization of the nuts. Proper identification of maturity level is critical, especially for commercial purposes. Early-stage coconuts are typically used for water, while mature brown coconuts are suitable for oil and copra production. Accurate classification can improve market pricing, harvest planning, and overall farm productivity.

Recent advancements in Artificial Intelligence (AI), particularly in deep learning and computer vision, have shown promise in automating agricultural tasks. Studies have employed Convolutional Neural Networks (CNNs) to classify fruits, detect crop diseases, and assess plant maturity using image data captured by drones or smartphones. In the context of coconut farming, integrating AI with drone technology can enable aerial image acquisition, providing a scalable solution to monitor coconut trees across large estates.

Several researchers have applied CNNs and YOLO (You Only Look Once) object detection models for identifying fruit ripeness in mangoes, apples, and bananas with high accuracy. Transfer learning techniques, such as using pre-trained models like ResNet and MobileNet, have further improved performance on small agricultural datasets. Moreover, deep learning models have also been implemented for yield estimation based on canopy cover, fruit count, and vegetative indices derived from aerial imagery.

In coconut-specific research, fewer works have addressed yield prediction and fruit maturity classification. One study used drone-captured images to estimate crown size and frond count, correlating them with coconut yield. However, it lacked the integration of maturity classification. Another work implemented an object detection model to detect coconut clusters, but it did not segment fruits by maturity stages.

To bridge this gap, our research proposes an **AI-powered drone system** that performs both **yield prediction** and **fruit maturity classification** (early, middle-aged, brown coconuts) using aerial imagery and deep learning techniques. This approach aims to deliver real-time, accurate insights that help farmers optimize harvest scheduling, improve crop monitoring, and enhance decision-making at scale.

## 1.2 Research Gap

The advancement of precision agriculture has led to the integration of deep learning models and drone imagery to automate and enhance tasks such as crop detection, health monitoring, and yield prediction. In this context, multiple studies have investigated coconut maturity detection using computer vision and artificial intelligence. For instance, the work by Novelero and Cruz [1] utilized UAV imagery with deep learning models to detect mature coconuts on trees. Their approach significantly improved fruit detection accuracy under varied lighting and occlusion conditions. However, the focus was limited to **on-tree fruit detection**, without incorporating multi-stage maturity levels or post-harvest predictions for yield estimation.

Bhaskar et al. [2] proposed a hybrid CNN-XGBoost model paired with sonographic techniques for autonomous coconut harvesting. While their innovation stands out in combining sensor data with deep learning, their study emphasized **mechanical harvesting systems**, rather than creating a mobile or drone-based real-time monitoring solution accessible to farmers or plantation managers. Furthermore, their system lacked spatial tracking of yield or disease-prone zones across plantations over time.

Several works have also attempted object detection and classification of coconut trees in plantation landscapes using deep learning. For example, Worachairungreung and Kulpanich [3] employed Faster R-CNN and Mask R-CNN for tree classification from UAV imagery. Their results demonstrated high accuracy in locating and classifying trees but **did not assess fruit maturity or yield potential**. Similarly, Karunarathna et al. [4] used YOLOv3 for mapping individual coconut trees; however, their research was confined to **tree detection and count estimation**, with no maturity-level differentiation or productivity analytics.

Mandava et al. [5] made progress in identifying the maturity of individual coconuts using deep learning models. Their classification was based on visual cues from close-up images of coconuts. However, the study relied on **static, non-aerial datasets**, limiting its applicability in large-scale farms or real-time scenarios. This constraint reduces scalability in commercial agriculture, especially for regions requiring continuous monitoring via drones or IoT-based solutions.

Moreover, Madhumathi et al. [6] proposed a drone-based monitoring system but emphasized tree health indicators and general plantation tracking. They did not delve into coconut fruit development stages or harvest forecasting. Teo et al. [7] presented an EfficientNet-based model for distinguishing between oil palm, coconut, and banana trees in UAV images. Though their model performed well in tree classification, **fruit-level analysis and predictive yield modeling were absent**.

In contrast to the above studies, this research aims to bridge the following key gaps:

1. **Multi-stage Maturity Detection** – Unlike previous studies that focus only on binary classification (mature/immature), this research introduces a multi-level classification model that detects the full coconut maturity spectrum (tender, semi-mature, mature) using aerial drone images.



2. **Yield Prediction Model** – While earlier works centered around detection, the proposed system integrates machine learning-based regression techniques to **predict potential yield** per tree/area, which is crucial for economic planning and supply chain optimization.
3. **Lightweight Deployment & Farmer Accessibility** – This study emphasizes the development of a **lightweight, mobile-accessible platform** powered by cloud-based processing, enabling farmers and plantation workers to gain insights without high-end hardware.
4. **Integration of Geolocation Data** – By mapping maturity and yield predictions to GPS-tagged imagery, the system provides **spatial tracking** of productivity, enabling better resource allocation and early detection of underperforming zones in plantations.

Therefore, this research contributes a novel, end-to-end solution that fills the missing link between coconut maturity identification, yield estimation, and actionable insights using deep learning and drone technology in an accessible format for large-scale agricultural use.

### 1.3 Research Problem

Coconut cultivation plays a vital role in the agricultural economy of tropical countries like Sri Lanka. Accurate yield prediction and maturity classification of coconuts are essential components for improving productivity, resource allocation, and timely harvesting. Traditionally, farmers estimate yield and coconut maturity based on visual observations and manual labor, which often leads to inconsistencies, inaccuracies, and inefficiencies in farm management and market planning.

Recent advancements in drone technology and artificial intelligence have enabled researchers to analyze aerial images for crop monitoring. Studies such as *On-tree Mature Coconut Fruit Detection Based on Deep Learning Using UAV Images* [1] have demonstrated that deep learning models like YOLO can identify mature coconuts using aerial imagery. However, these works often focus only on the detection of **fully matured coconuts**, ignoring the significance of **classifying coconuts into different maturity stages** (early, middle-aged, and brown) which is crucial for optimized harvesting schedules and processing needs.

Furthermore, research such as *Automated Coconut Tree Counting and Fruit Maturity Analysis Using Deep Learning* [2] has concentrated on estimating the number of fruits, yet lacks a **predictive model for future yield** based on temporal or environmental variables such as historical weather patterns, flowering stages, or nutrient levels. This highlights a clear research gap in **integrating yield prediction with maturity classification using drone imagery and AI-based models**.

Additionally, existing approaches often depend on expensive hardware setups or cloud-based systems, which are not accessible to most small-scale coconut farmers in rural Sri Lanka. There is also limited research that combines **multi-spectral drone data with machine learning regression models** (e.g., Random Forest, LSTM) to forecast yield while classifying fruit stages simultaneously. This dual-purpose system could drastically improve harvest planning and reduce post-harvest losses due to improper timing.

Therefore, this research aims to fill these gaps by developing an AI-based system that integrates UAV imagery with deep learning models to:

1. Predict the yield of coconut trees over time.
2. Classify coconuts based on their maturity stage (early, middle-aged, brown).
3. Deliver this functionality through a user-friendly, cost-effective platform accessible to local farmers.

This solution will empower farmers with **data-driven decision-making**, enabling better planning, timely harvesting, and maximizing market value. By focusing on localized data and real-time drone monitoring, the proposed system also addresses limitations in scalability and generalizability found in current literature.

## 2. Objectives

### 2.1 Main Objectives

The main objective of this study is to develop an automated system to classify the maturity level of coconuts and predict yield using image processing and deep learning techniques. The proposed solution will assist farmers in making timely decisions on harvesting, ultimately increasing productivity and minimizing wastage. And to develop an intelligent AI-powered system that accurately predicts coconut yield and classifies coconut maturity stages (early, middle-aged, brown) using drone-captured imagery and deep learning techniques, enabling farmers to make data-driven decisions for optimal harvesting and resource allocation.

### 2.2 Specific Objectives

#### ☐ Coconut Detection and Localization

- Develop a deep learning model to automatically detect and locate coconuts in drone-captured aerial images
- Achieve minimum 90% accuracy in coconut identification across various lighting and environmental conditions

#### ☐ Maturity Stage Classification

- Implement a CNN-based classification system to categorize coconuts into three maturity stages:
  - Early stage (tender/green coconuts)
  - Middle-aged stage (semi-mature coconuts)
  - Brown stage (fully mature coconuts)
- Achieve classification accuracy of at least 85% for each maturity category

#### ☐ Yield Prediction Model

- Develop a regression model to predict coconut yield per tree and per plantation area
- Integrate temporal data, crown characteristics, and fruit count for accurate forecasting
- Provide yield predictions with confidence intervals for harvest planning

#### ☐ Real-time Processing and Integration

- Create a system capable of processing drone imagery in near real-time
- Integrate with the broader AI-powered drone platform for comprehensive plantation monitoring
- Develop user-friendly interfaces for farmers and agricultural officers

### 3. Methodology

The proposed system is designed to assist coconut farmers by automating the process of identifying coconut maturity levels and predicting the potential yield using image processing and machine learning techniques. The system allows users to capture images of coconut trees, which are then analyzed to classify the maturity stages (immature, mature, overripe) using a deep learning model. Once the coconuts are classified, the system predicts the estimated yield based on historical data, tree count, and classified maturity level.

The output of the system helps farmers to plan the harvest effectively, reduce losses, and improve decision-making regarding resource allocation, workforce planning, and market forecasting. All data is stored in a centralized database and can be accessed through a cross-platform mobile application.

#### 3.1 System Architecture

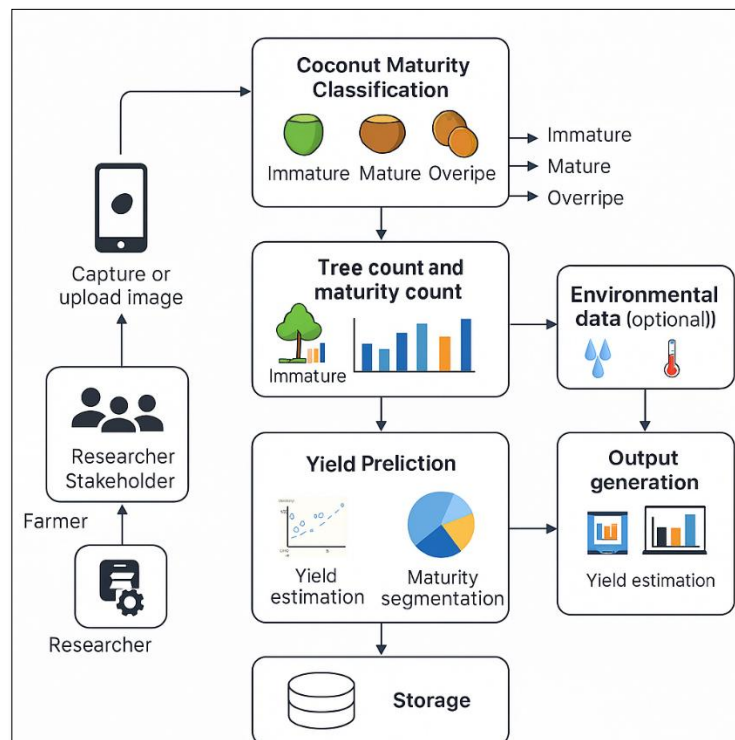


Figure 3.1: Overall System Diagram

**Figure 3.1** presents the system architecture of the proposed coconut maturity and yield prediction component. The system is designed to allow users to capture or upload images of coconuts through a mobile application interface. These images are then converted and processed using a deep learning model, specifically trained to identify the maturity stage of the coconut. Based on the maturity detection results and additional environmental parameters such as humidity, temperature, and rainfall, a machine learning model is applied to predict the expected yield. The final results are stored and visualized through a centralized database, enabling farmers and stakeholders to make informed harvesting and market decisions.

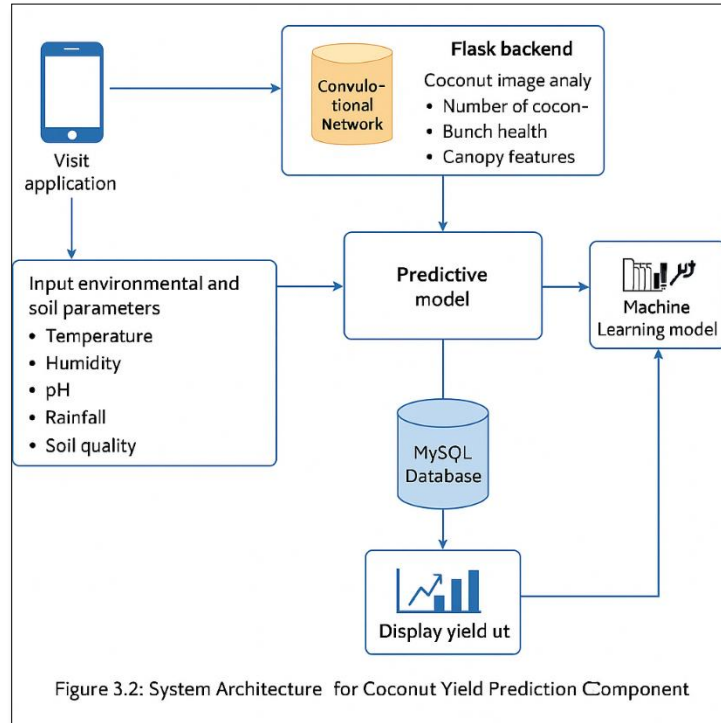


Figure 3.2: System Architecture for Coconut Yield Prediction Component

Figure 3.2 illustrates the overall high-level architecture of the proposed coconut yield prediction component. This system focuses on predicting the yield of coconut trees by analyzing visual data of the tree along with associated environmental parameters.

Initially, the user (such as a farmer or stakeholder) accesses the cross-platform mobile application built using **React Native**. Through the application, users can either capture a real-time image of the coconut tree or upload an existing image. Once the image is captured or uploaded, it is converted into a **Base64** string and transmitted to the **Flask backend server** for processing.

The Flask server includes a trained **Convolutional Neural Network (CNN)** model that analyzes the image to determine the **number of visible coconuts**, the **health status of the bunch**, and **canopy features**. In parallel, user-provided environmental and soil parameters such as **temperature, humidity, pH, rainfall, and soil quality** are input into the **Machine Learning model** (e.g., Random Forest or XGBoost) that has been trained using historical datasets.

These two data streams — image analysis and environmental data — are then combined in a **predictive model** which estimates the **expected yield output (number of coconuts per tree)**. The results are stored in the

**MySQL database** and displayed through the app, allowing the user to view current and historical predictions, and take action accordingly (e.g., plan harvesting or adjust care).

| Category      | Details                                                                     |
|---------------|-----------------------------------------------------------------------------|
| Technologies  | Python, TensorFlow, React Native, Flask, MySQL, OpenCV, Keras               |
| Techniques    | Machine Learning, Image Processing, Feature Engineering, Data Normalization |
| Algorithms    | CNN, Random Forest, XGBoost                                                 |
| Architectures | ResNet-50, VGG-19, Inceptionv3                                              |

Table 3.1: Technologies, Techniques, Algorithms, Architectures Used

3.1.1 Software Solution

The development of the coconut maturity and yield prediction system follows the **Software Development Life Cycle (SDLC)** to ensure structured and high-quality software delivery. SDLC helps the development team plan, design, implement, test, and maintain the system efficiently. By applying this process, potential risks are minimized early, ensuring alignment with user needs and expectations.

Given the dynamic nature of user feedback and agricultural variability, the **Agile methodology** was adopted. Agile promotes iterative development, continuous feedback, and flexibility, making it easier to adapt to changes during development. Among Agile practices, **Scrum** was chosen due to its simplicity and strong team collaboration features. The Scrum framework enables quick development cycles (sprints), allowing for incremental delivery of key features like image capturing, maturity detection, and yield prediction.

This adaptive approach ensures the mobile application remains reliable, scalable, and responsive to stakeholder requirements in real-world agricultural environments.

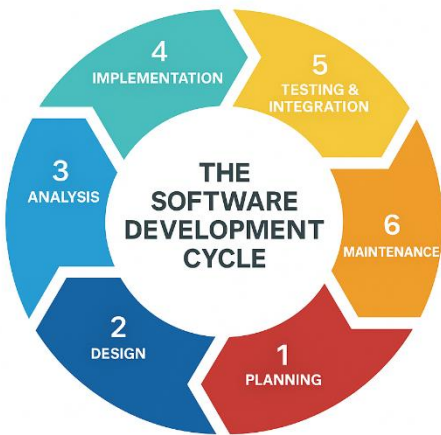


Figure 3.3: Agile Software Development Cycle

- Requirement Gathering

- Primary Data Collection

- Field Surveys

- Manual data collection was conducted in collaboration with the **Lunuwila Coconut Research Institute** and 10–15 selected coconut plantations across different regions of Sri Lanka. These plantations were chosen based on their diversity in soil, climate, and cultivation practices to ensure comprehensive data representation.

- Image Dataset

- Over **1000 high-resolution images** of coconut trees planning to capture using drones (UAVs) at multiple angles and elevations. These images cover various maturity stages of coconuts to support classification and yield prediction model training.

- Ground Truth Data

- Expert agricultural officers from Lunuwila manually **classified and counted** the coconuts based on maturity stages to label the image dataset. This ground-truth labeling serves as the foundation for supervised learning.

- Temporal Data

- Monitoring was conducted **monthly over multiple seasons** to record actual harvest data and correlate it with image-based predictions. This allowed the collection of temporal trends in yield and maturity.

- Data Collection Protocol

- Image Capture Standards

- Using drones equipped with minimum 4K resolution cameras.
      - Captured multiple angles: top-down, side view (45°), and ground level.
      - Ensured consistent lighting by conducting flights during early mornings.
      - Embedded **GPS coordinates and timestamp metadata** for each image.

- Ground Validation

- Agricultural experts **manually classified** coconut maturity into stages.
      - Physical measurements of randomly selected coconuts for **size and weight validation**.
      - **Actual harvest data** (yield per tree) was recorded to validate the system's prediction output.

- Conducting a Survey

A survey was conducted targeting farmers, agricultural officers, and other stakeholders to identify current challenges in predicting coconut yield and maturity. This helped assess the awareness level, technological adaptation, and expectations regarding a digital prediction system.

- Feasibility Study (Planning)

- Economic Feasibility

The proposed system is designed to be deployed as a low-cost mobile/web-based platform that utilizes existing smartphone hardware for image acquisition. This significantly reduces implementation costs. By leveraging open-source tools and pre-trained models where possible, the development cost is minimized while ensuring reliable output.

- Scheduled Feasibility

This component is designed to be completed within the allocated time frame of the main project schedule. The scope of the component, including model training, validation, and integration with the interface, is feasible within the proposed timeline without delays.

- Technical Feasibility

The solution will be developed using tools such as Python, TensorFlow, and OpenCV for image processing and deep learning. It requires expertise in machine learning model training, dataset handling, and full-stack development using technologies like React Native or Flutter. The team possesses the required technical knowledge to complete the proposed system successfully.

- Design (System and Software Design Document)

Following the requirement gathering and feasibility study, the design phase involves planning the overall architecture of the yield prediction module. This includes the data flow for input image processing, coconut maturity detection, and yield prediction using ML models. Detailed software design documents such as DFDs, UML diagrams, and architectural flowcharts will be prepared to guide the development phase.



## Sequence Diagram

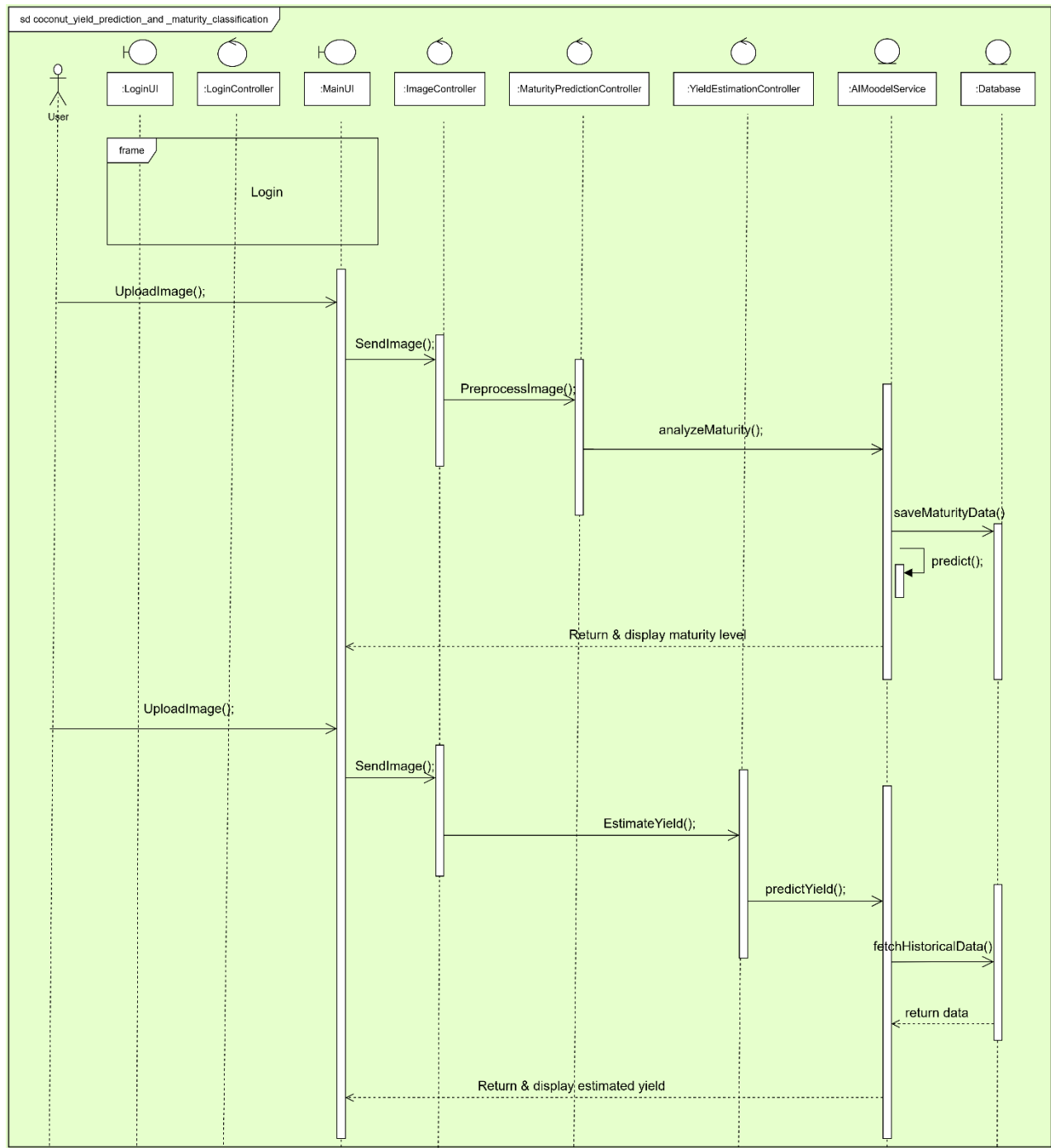


Figure 3.4: Sequence Diagram of Coconut Yield Prediction and Maturity Classification

Figure 3.4 sequence diagram illustrates the interaction between various system components during the processes of coconut maturity classification and yield estimation. The user initiates the system by uploading an image, which is first processed by the **ImageController**. For maturity prediction, the image is forwarded to the **MaturityPredictionController** and analyzed by the **AIModelService**, which internally invokes its prediction logic and stores the results in the database. In the yield estimation phase, the same or another image is processed and sent to the **YieldEstimationController**, which again utilizes the AI model service. The service fetches historical data from the database and performs prediction. The predicted maturity level and yield values are then

returned to the user interface for display. This diagram demonstrates the modular architecture, asynchronous interactions, and encapsulated AI logic essential for system maintainability and scalability.

## Use Case Diagram



Figure 3.5: Use Case Diagram of Coconut Yield Prediction and Maturity Classification

This use case diagram illustrates the functional interactions between the system and its primary actors: the **Coconut Farmer** and the **System Administrator**. The diagram represents the core functionalities of the system, focusing on two key components—**maturity level classification** and **yield estimation** based on uploaded coconut tree images.

The **Coconut Farmer** interacts with the system to register, log in, capture images of coconut trees, and upload them for processing. The system includes the functionality to validate user credentials during login, and it also handles error messages if credentials are invalid. Once the image is uploaded, the farmer can initiate the **maturity classification**, which internally includes image analysis and saving the maturity data. For **yield estimation**, the system includes steps to fetch historical data and predict the expected yield based on trained models.

The **System Administrator**, on the other hand, has access to system-wide operations such as viewing prediction results, generating reports, and performing maintenance tasks to ensure the system remains functional and up to date.

This diagram helps identify all key user interactions and internal processes, offering a clear overview of the scope and responsibilities of each system component.

- **Implementation (Development)**

Below are the sub-functionalities that will be implemented in this development phase for the Coconut Maturity and Yield Prediction component:

- **Implementation of a mobile application** using React Native to allow users to capture or upload images of coconuts.
- **Prediction of coconut maturity stages** using a trained deep learning model (e.g., CNN or YOLOv5) based on the uploaded drone imagery or mobile-captured photos.
- **Prediction of coconut yield quantity** using a machine learning model (e.g., Random Forest or XGBoost), trained on historical harvest data, maturity distribution, and environmental parameters.
- **Visualization and reporting** of predicted results through a user-friendly interface, providing users insights into current maturity levels and expected yield.
- **Data storage and processing** using Firebase/Flask backend and a centralized database (e.g., MySQL) for secure and scalable access.

- **Testing**

Testing will ensure that the system is robust, functional, and reliable across all use cases. Testing procedures include:

- **Unit Testing** – To verify the correctness of individual components (e.g., image classification module, prediction module).
- **Integration Testing** – To ensure different modules (frontend, backend, prediction engine) work together as expected.
- **System Testing** – To test the entire application workflow from image upload to yield report generation.
- **Acceptance Testing** – To verify that the system meets all functional and non-functional requirements as expected by the stakeholders and end-users.

### 3.1.2 Commercialization

This research proposes the development of a mobile application designed to assist coconut farmers and plantation stakeholders in assessing coconut maturity stages and predicting harvest yield. The intended users of the system include coconut farmers, agricultural researchers, industry stakeholders, and supply chain entities.

The system will be made available in two versions: a **Basic (Commodity) Version** and a **Premium Version**.

The application will be available in two deployment models:

- A Basic Free Version
- A Premium Subscription Version

The Basic Version will provide:

- Coconut maturity detection from uploaded or drone-captured images
- A visual report on maturity stage distributions
- Basic recommendations for harvest scheduling

The Premium Version will offer enhanced capabilities including:

- Forecasting of coconut yield using time-series analysis and image classification
- Integration with drone services for automated maturity scanning
- Historical yield trend visualization for plantation planning
- Environmental factor-based maturity risk alerts
- Personalized harvest recommendations and expert support

In the Sri Lankan context, few mobile tools specifically address coconut maturity tracking. Our application is designed to be lightweight and accessible for users with basic technical skills. It is especially suitable for coconut farmers aged between 25–65, and requires no prior knowledge of machine learning or AI.

The project team consists of individuals with academic expertise in computer science, AI, and agricultural informatics. Members have hands-on experience working with local institutions such as the Coconut Research Institute in Lunuwila, ensuring domain relevance.

Monetization will be based on subscription tiers, with additional offerings like institutional licenses, yield consultancy, and drone-data integrations for large-scale plantation operations. The product will be promoted via agricultural exhibitions, CRI outreach programs, local media coverage, and collaborations with agricultural officers.

### 3.1.3 Future Scope

For future enhancements, the coconut maturity detection application is envisioned to evolve into a comprehensive smart agriculture solution. One key improvement will be the integration of **automated drone systems** to perform scheduled aerial image collection across coconut plantations, eliminating the need for manual operation and ensuring consistent data acquisition over time.

To support real-time decision-making, the application can be expanded to integrate with **IoT-based weather monitoring systems**, allowing it to correlate coconut maturity stages with environmental conditions such as rainfall, humidity, and temperature. This data will aid in **predictive modeling** of harvest cycles, improving yield forecasting accuracy.

Another potential advancement is the **incorporation of additional coconut maturity indicators**, such as nut sound analysis and size estimation using 3D imaging, to increase the accuracy of maturity classification. Furthermore, the application could support **multilingual user interfaces** to promote usability among regional farming communities across Sri Lanka.

In the long term, the system could be linked with **automated harvesting tools** or **smart notification systems** that alert farmers of optimal harvest windows. This would not only streamline operations but also reduce harvest losses due to premature or delayed coconut collection. Finally, cloud integration and a farmer dashboard can be introduced to offer **historical data analysis**, **plantation performance tracking**, and **automated reporting** features for stakeholders and plantation managers.

## 4. Project Requirements

### 4.1 Functional Requirements

1. The system should allow users to capture or upload high-resolution images of coconut bunches using a smartphone or drone-based camera.
2. The system should classify the maturity stage of coconuts (e.g., **immature**, **semi-mature**, **mature**) using a trained AI model.
3. The system should estimate the **number of coconuts per tree** using object detection techniques.
4. The system should predict the **expected yield per tree** or per area based on image analysis and environmental data (e.g., past harvests, growth stage).
5. The system should allow users to input contextual parameters (e.g., age of tree, region, date of planting) for more accurate predictions.
6. The system should store and manage historical data (image, prediction results, tree ID) for temporal yield analysis.
7. The system should generate visual reports or dashboards with prediction summaries for farm management decisions.

### 4.2 Non-functional Requirements

1. **User-Friendliness** – The application should be easy to use for farmers and agricultural officers, with a clean, intuitive interface.
2. **Accuracy** – The coconut count and maturity classification should achieve **over 90% accuracy**, validated with expert ground-truth data.
3. **Performance** – The system should return maturity classification and yield prediction results within **20–30 seconds** per image.
4. **Availability** – The mobile application should work both online and offline, syncing data once connectivity is available.
5. **Scalability** – The system must handle increasing data from multiple plantations and support simultaneous users.
6. **Security** – All user data and farm images must be securely stored and accessible only by authorized users.
7. **Cross-platform Compatibility** – The app should be deployable on both **Android and iOS** platforms using a shared codebase (e.g., React Native).

### 4.3 System Requirements

1. **VS Code** – To develop and manage the source code using Python.
2. **Flask Server** – To deploy and run the trained maturity classification and yield prediction models.
3. **React Native** – To build the cross-platform mobile application for farmers and plantation managers.
4. **Python** – For backend development and machine learning model implementation.
5. **TensorFlow / PyTorch** – For training and deploying the image classification and regression models.

### 4.4 User Requirements

This mobile application will be used by the following user types:

1. **Coconut Farmers:**  
Farmers can use the application to upload or capture images of coconut trees. Based on the maturity classification, they can determine the optimal harvesting time and estimate expected yield per tree or per area. This helps them make informed harvesting and labor allocation decisions.
2. **Agricultural Researchers:**  
Researchers can use the platform to access annotated coconut imagery datasets and prediction results for research on maturity trends, yield forecasting models, and plantation health. This supports further agricultural innovation and model improvements.
3. **Plantation Managers / Government Officers:**  
Managers can monitor large plantations using drone-integrated data, access dashboard visualizations of coconut maturity stages, and plan logistics, export schedules, or training programs based on yield forecasts.

# 5. Gantt Chart

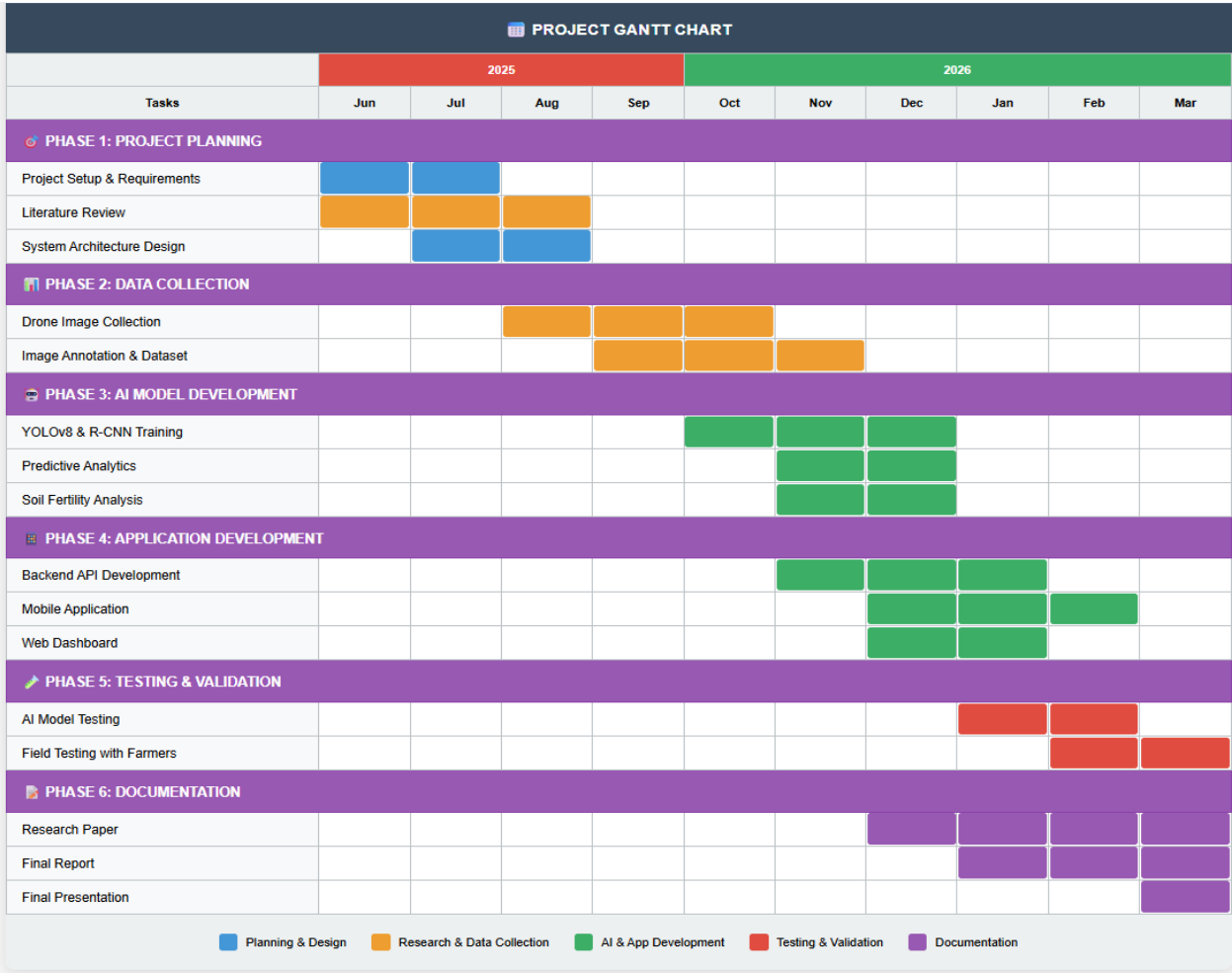


Figure 5.1: Gantt Chart

The Gantt chart illustrates the comprehensive project timeline for the coconut maturity classification and yield prediction system, spanning from **June 2025 to March 2026**. The project is divided into six key phases, each with distinct deliverables and dependencies:

- 1. Phase 1: Project Planning
  - **Duration:** June to August 2025  
This phase involves foundational work including project setup, requirement analysis, literature review, and system architecture design. It establishes the project scope, objectives, and technical roadmap.
- 2. Phase 2: Data Collection
  - **Duration:** August to November 2025



Data collection focuses on drone image acquisition from coconut plantations and subsequent annotation for maturity stages. This phase ensures a high-quality, labeled dataset essential for AI model training.

3. Phase 3: AI Model Development

- **Duration:** October 2025 to January 2026

This phase covers training object detection models (YOLOv8, R-CNN) for fruit detection, implementing yield prediction algorithms, and analyzing soil fertility correlations. It's the core of the intelligent system development.

4. Phase 4: Application Development

- **Duration:** November 2025 to February 2026

Backend APIs, the cross-platform mobile application, and the web-based dashboard are developed during this phase. These components integrate the trained AI models and deliver the insights to end users.

5. Phase 5: Testing & Validation

- **Duration:** January to March 2026

Model performance and system functionality are tested under lab conditions and validated in the field with real farmer input. Feedback from this phase ensures system robustness and reliability.

6. Phase 6: Documentation

- **Duration:** December 2025 to March 2026

Research paper drafting, final project report preparation, and presentation creation are conducted to communicate project outcomes and research contributions.

Each phase is color-coded for clarity, aligning tasks with their respective categories:

- **Blue:** Planning & Design
- **Orange:** Research & Data Collection
- **Green:** AI & App Development
- **Red:** Testing & Validation
- **Purple:** Documentation

# 5.1 Work Breakdown Structure (WBS)

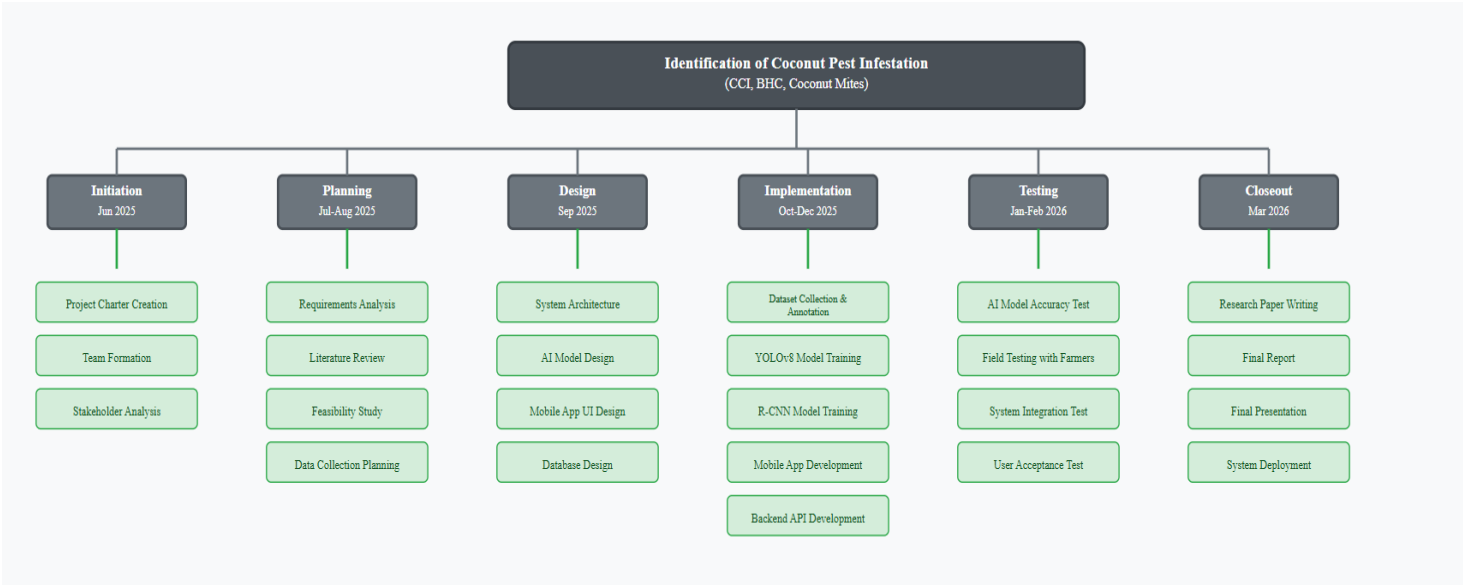


Figure 5.2: Work Breakdown Chart of Coconut Yield Prediction and Maturity Classification

The **Work Breakdown Structure (WBS)** in Figure 5.2 outlines the hierarchical decomposition of tasks involved in the development of the **Coconut Yield Prediction and Maturity Classification System**. The chart is divided into six primary phases, each representing a critical milestone in the system development life cycle:

1. Initiation (June 2025)  
This phase focuses on establishing the foundation of the project:
  - **Project Charter Creation:** Defines project objectives, scope, and stakeholders.
  - **Team Formation:** Assigns roles and responsibilities.
  - **Stakeholder Analysis:** Identifies key stakeholders and their expectations.
2. Planning (July–August 2025)  
Detailed planning is conducted to ensure smooth project execution:
  - **Requirement Analysis & Feasibility Study:** Determines functional and technical needs.
  - **Literature Review:** Assesses existing research and solutions.
  - **Data Collection Planning:** Prepares for field data acquisition via drones.
3. Design (September 2025)  
System design is formulated for both AI and application aspects:
  - **System Architecture:** Defines high-level system structure.
  - **AI Model Design:** Outlines model types for fruit maturity and yield prediction.
  - **Mobile App UI & Dashboard Design:** Plans user-friendly interfaces for farmers and stakeholders.
4. Implementation (October–December 2025)

Development and integration of system components begin:

- **Dataset Collection & Annotation:** Gathering high-resolution images and labeling maturity stages.
- **Model Training (YOLOv8 & R-CNN):** Train deep learning models for fruit detection and yield estimation.
- **Mobile App & Backend Development:** Build cross-platform app and API to host and communicate with the models.

5. Testing (January–February 2026)

Validation ensures system performance and usability:

- **Model Accuracy Tests:** Evaluate prediction reliability.
- **Field Testing with Farmers:** Validate the system in real-world scenarios.
- **System & User Acceptance Testing:** Ensure functional completeness and user satisfaction.

6. Closeout (March 2026)

Finalization of the project and deliverables:

- **Research Paper & Final Report Writing:** Document technical and research contributions.
- **Final Presentation:** Communicate results to stakeholders.
- **System Deployment:** Launch the application for actual usage.

## 6. Budget and Budget Justification

| Requirement                                          | Cost (Rs.)        |
|------------------------------------------------------|-------------------|
| Travelling cost for field visits and data collection | 10,000.00         |
| Data Collection (drone and equipment)                | 45,000.00         |
| Cloud hosting and AI model training                  | 25,000.00         |
| Mobile app development and deployment                | 15,000.00         |
| App Store hosting and maintenance                    | 12,000.00         |
| Research documentation and publication               | 18,000.00         |
| <b>Total Expenses</b>                                | <b>135,000.00</b> |

Table 6.1: Expenses for the proposed system

The total estimated budget for the implementation of the **Coconut Yield Prediction and Maturity Classification System** is **Rs. 135,000.00**, allocated across six key components to ensure successful execution of the project. Below is the justification for each budget item:

1. Travelling Cost for Field Visits and Data Collection – Rs. 10,000.00  
This allocation covers transportation and logistical expenses for the research team to visit coconut plantations across various regions. These visits are essential for collecting real-world data and validating field observations.
2. Data Collection (Drone and Equipment) – Rs. 45,000.00  
This is the largest budget item, justified by the need for **high-quality drone imagery** to capture coconut trees at various maturity stages. The cost includes renting or purchasing drone equipment, image capturing accessories, and related tools.
3. Cloud Hosting and AI Model Training – Rs. 25,000.00  
This budget supports **cloud-based infrastructure** (e.g., AWS, Google Cloud) to host the dataset and train deep learning models. It ensures scalable, secure, and efficient training of the yield prediction and maturity classification models.
4. Mobile App Development and Deployment – Rs. 15,000.00  
Funds allocated here will support the **development of a cross-platform mobile application** using frameworks like React Native. It also includes initial testing and deployment efforts.

5. App Store Hosting and Maintenance – Rs. 12,000.00  
This covers **annual charges for publishing the mobile app** on platforms like Google Play Store and Apple App Store, along with basic maintenance costs to ensure ongoing availability and updates.
6. Research Documentation and Publication – Rs. 18,000.00  
This budget supports the **production of research reports, final documentation, and publishing expenses**, ensuring that the outcomes are formally recorded and disseminated to relevant stakeholders and academia.
7. Total Budget: Rs. 135,000.00  
This budget plan reflects a balanced and practical distribution of costs across data acquisition, development, deployment, and documentation, supporting both technical implementation and research dissemination.

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